

Practice Questions for 2024-11-21-EconNetSimplex

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1 Questions

1. In the economic interpretation of the network simplex method (Slides 1-6), what is the significance of the equation $y_i + c_{ij} = y_j$ in the first step, where y_i is the price at node i , c_{ij} is the cost from i to j , and y_j is the price at node j ?
 - a. It ensures that the total cost of shipping from node i to node j is minimized.
 - b. It guarantees that no arbitrage opportunities exist in the network.
 - c. It determines the optimal flow along each arc in the network.
 - d. It calculates the reduced cost for each arc in the network.
 - e. It establishes the initial feasible solution for the network simplex method.
2. In the second step of the network simplex method's economic interpretation (Slide 5), the algorithm checks the condition $y_i + c_{ij} < y_j$. What is the economic significance of this inequality?
 - a. It identifies an arc where increasing the flow will decrease the total cost.
 - b. It indicates an arc where the current prices allow for profitable arbitrage.
 - c. It determines the leaving arc in the current iteration of the simplex method.
 - d. It signifies that the current solution is optimal.
 - e. It calculates the reduced cost of the arc (i, j) .
3. In the third step of the network simplex method's economic interpretation (Slide 6), the algorithm adjusts the flow to maintain feasibility. What is the economic interpretation of the equation $\tilde{y}_j - \tilde{y}_i = c_{ij}$ after the adjustment?
 - a. It ensures that the total flow in the network is conserved.
 - b. It guarantees that the new flow is optimal.
 - c. It establishes a new set of equilibrium prices in the network.
 - d. It calculates the change in the objective function value.
 - e. It identifies the next entering arc for the next iteration.
4. The network simplex method uses a spanning tree to represent a feasible solution. Why is starting with a feasible solution on a spanning tree (Slide 2) advantageous in the context of the economic interpretation?

- a. It simplifies the calculation of the reduced costs.
 - b. It ensures that the initial solution is optimal.
 - c. It allows for easy identification of arbitrage opportunities.
 - d. It guarantees that the flow is conserved at each node.
 - e. It provides a straightforward way to establish a set of initial prices that prevent arbitrage.
5. In the economic interpretation of the network simplex method (Slide 4), what is the significance of the equation $y_i + c_{ij} = y_j$ for an arc (i, j) in the spanning tree T ?
 - a. It ensures that the flow along arc (i, j) is maximized.
 - b. It guarantees that the total cost of the network flow is minimized.
 - c. It establishes a set of "fair prices" such that there is no incentive for arbitrage between nodes i and j .
 - d. It determines the optimal flow along each arc in the network.
 - e. It ensures that the supply at node i equals the demand at node j .
6. In Slide 5, the second step of the network simplex method involves checking the condition $y_i + c_{ij} < y_j$ for arcs (i, j) not in the spanning tree T . What is the economic interpretation of this inequality?
 - a. It indicates that the current flow is optimal.
 - b. It suggests an opportunity for a competitor to profitably ship goods from node i to node j .
 - c. It means the arc (i, j) should be removed from the network.
 - d. It implies that the prices at nodes i and j are inconsistent.
 - e. It signifies that the supply at node i exceeds the demand at node j .
7. In the third step of the network simplex method's economic interpretation (Slide 6), the flow is adjusted to maintain feasibility. What is the economic significance of the equation $\tilde{y}_j - \tilde{y}_i = c_{ij}$ after the adjustment, where $\tilde{y}_i$ and $\tilde{y}_j$ are the adjusted prices at nodes i and j , respectively?
 - a. It ensures that the total flow in the network is conserved.
 - b. It guarantees that the total cost of the network flow is minimized.
 - c. It establishes a new set of "fair prices" where there is no longer an incentive for arbitrage between nodes i and j along the newly added arc.
 - d. It indicates that the flow along arc (i, j) is at its maximum capacity.
 - e. It implies that the supply at node i is equal to the demand at node j .
8. Why is starting with a feasible solution on a spanning tree (Slide 2) advantageous in the context of the economic interpretation of the network simplex method?
 - a. It simplifies the calculation of the maximum flow in the network.
 - b. It ensures that the initial solution satisfies all supply and demand constraints.

- c. It allows for a straightforward determination of "fair prices" using the equation $y_i + c_{ij} = y_j$ for arcs in the tree.
- d. It guarantees that the algorithm will converge to the optimal solution in a single iteration.
- e. It makes it easier to identify the minimum cut in the network.

2 Answers

1. In the economic interpretation of the network simplex method (Slides 1-6), what is the significance of the equation $y_i + c_{ij} = y_j$ in the first step, where y_i is the price at node i , c_{ij} is the cost from i to j , and y_j is the price at node j ?
 - a. While minimizing shipping costs is a goal, this equation doesn't directly minimize the total cost. The total cost is minimized through the entire simplex process, not just this step.
 - b. The equation $y_i + c_{ij} = y_j$ ensures that there are no arbitrage opportunities. If the equation didn't hold, one could buy at a lower price at node i , ship to node j , and sell at a higher price, making a profit without any risk. This condition is crucial for establishing a set of "fair prices" in the network.
 - c. The optimal flow is determined through the entire simplex process, not just by this equation. This equation establishes a condition for fair pricing, which is a step in finding the optimal flow.
 - d. Reduced costs are calculated later in the simplex method. This equation is about establishing fair prices based on shipping costs.
 - e. This equation is a condition for the prices, not the initial feasible solution itself. The initial feasible solution is established by selecting a spanning tree.
2. In the second step of the network simplex method's economic interpretation (Slide 5), the algorithm checks the condition $y_i + c_{ij} < y_j$. What is the economic significance of this inequality?
 - a. While it identifies an arc for improvement, it doesn't directly relate to decreasing the total cost. The total cost is minimized through the entire simplex process.
 - b. The inequality $y_i + c_{ij} < y_j$ signifies that it is profitable to buy at node i , ship along arc (i, j) , and sell at node j . This represents an arbitrage opportunity, indicating that the current prices are not in equilibrium and the solution is not optimal.
 - c. The leaving arc is determined later in the iteration, based on the flow adjustments needed to maintain feasibility.
 - d. This inequality indicates that the current solution is *not* optimal, as there is a profitable arbitrage opportunity.
 - e. The reduced cost is related to this inequality but is not directly calculated by it. The reduced cost is implicitly considered in the comparison.
3. In the third step of the network simplex method's economic interpretation (Slide 6), the algorithm adjusts the flow to maintain feasibility. What is the economic interpretation of the equation $\tilde{y}_j - \tilde{y}_i = c_{ij}$ after the adjustment?
 - a. Flow conservation is maintained throughout the process, not just by this equation. This equation focuses on price equilibrium.
 - b. Optimality is not guaranteed after this step; further iterations may be needed.
 - c. After adjusting the flow, the equation $\tilde{y}_j - \tilde{y}_i = c_{ij}$ establishes a new set of equilibrium prices. The prices are adjusted to reflect the new flow, eliminating the arbitrage opportunity identified in the previous step.
 - d. The change in the objective function value is a consequence of the flow adjustment, not directly calculated by this equation.
 - e. The next entering arc is identified in the next iteration, not in this step.

4. The network simplex method uses a spanning tree to represent a feasible solution. Why is starting with a feasible solution on a spanning tree (Slide 2) advantageous in the context of the economic interpretation?
 - a. Reduced costs are calculated later in the algorithm, not directly from the spanning tree.
 - b. The initial solution on a spanning tree is feasible but not necessarily optimal.
 - c. While the spanning tree helps, the identification of arbitrage opportunities is done by checking the condition $y_i + c_{ij} < y_j$, not solely from the tree structure.
 - d. Flow conservation is a constraint of the problem, and the spanning tree helps maintain feasibility, but it doesn't directly guarantee conservation.
 - e. A spanning tree provides a simple structure where each arc has a unique path from the source to the sink. This allows for a straightforward calculation of prices that prevent arbitrage, forming the basis for the economic interpretation of the algorithm. The initial prices are set such that $y_i + c_{ij} = y_j$ for each arc in the spanning tree.

5. In the economic interpretation of the network simplex method (Slide 4), what is the significance of the equation $y_i + c_{ij} = y_j$ for an arc (i, j) in the spanning tree T ?
 - a. The equation does not directly maximize flow; that's determined by the overall optimization process.
 - b. Minimizing total cost is the overall goal, but this equation is a condition for a specific part of the solution.
 - c. The equation $y_i + c_{ij} = y_j$ represents a condition of no arbitrage. If the equation didn't hold, a competitor could buy at node i and sell at node j for a profit, contradicting the "fair price" assumption.
 - d. The optimal flow is determined by the entire simplex algorithm, not just this single equation.
 - e. This equation doesn't directly relate supply and demand; those are represented by the node labels b_i and b_j .

6. In Slide 5, the second step of the network simplex method involves checking the condition $y_i + c_{ij} < y_j$ for arcs (i, j) not in the spanning tree T . What is the economic interpretation of this inequality?
 - a. The inequality indicates the opposite: the current flow is *not* optimal.
 - b. If $y_i + c_{ij} < y_j$, it means a competitor could profitably buy at node i , ship to node j , and sell at a higher price than the current market price at node j , indicating a non-optimal solution.
 - c. The arc is not removed; it's considered for inclusion in the spanning tree to improve the solution.
 - d. The prices are not necessarily inconsistent; the inequality highlights an opportunity for improvement.
 - e. The inequality doesn't directly relate to supply and demand; those are represented by the node labels.

7. In the third step of the network simplex method's economic interpretation (Slide 6), the flow is adjusted to maintain feasibility. What is the economic significance of the equation $\tilde{y}_j - \tilde{y}_i = c_{ij}$ after the adjustment, where $\tilde{y}_i$ and $\tilde{y}_j$ are the adjusted prices at nodes i and j , respectively?
 - a. Flow conservation is maintained throughout the algorithm, but this equation specifically addresses pricing.

- b. Cost minimization is the overall goal, but this equation focuses on price equilibrium.
 - c. After adjustment, the equation $\tilde{y}_j - \tilde{y}_i = c_{ij}$ ensures that the adjusted prices eliminate the arbitrage opportunity identified in step 2. The new prices reflect the cost of shipping between the nodes.
 - d. The equation doesn't directly relate to capacity; that's a separate constraint.
 - e. Supply and demand are represented by node labels, not this equation.
8. Why is starting with a feasible solution on a spanning tree (Slide 2) advantageous in the context of the economic interpretation of the network simplex method?
- a. Maximum flow is not directly related to the initial feasible solution on a spanning tree.
 - b. While the initial solution must be feasible, starting on a spanning tree doesn't guarantee this; it's a choice to simplify the initial price calculation.
 - c. A spanning tree provides a connected subgraph without cycles. This structure allows for a direct calculation of prices using the equation $y_i + c_{ij} = y_j$ for each arc in the tree, providing a starting point for the economic interpretation.
 - d. The algorithm typically requires multiple iterations to reach optimality.
 - e. Minimum cut is a concept related to the max-flow min-cut theorem, not directly to the initial feasible solution.